

GPU image rotation and responsible HPC for AI

CUDA design and performance • Evidence, trade-offs and accountable use

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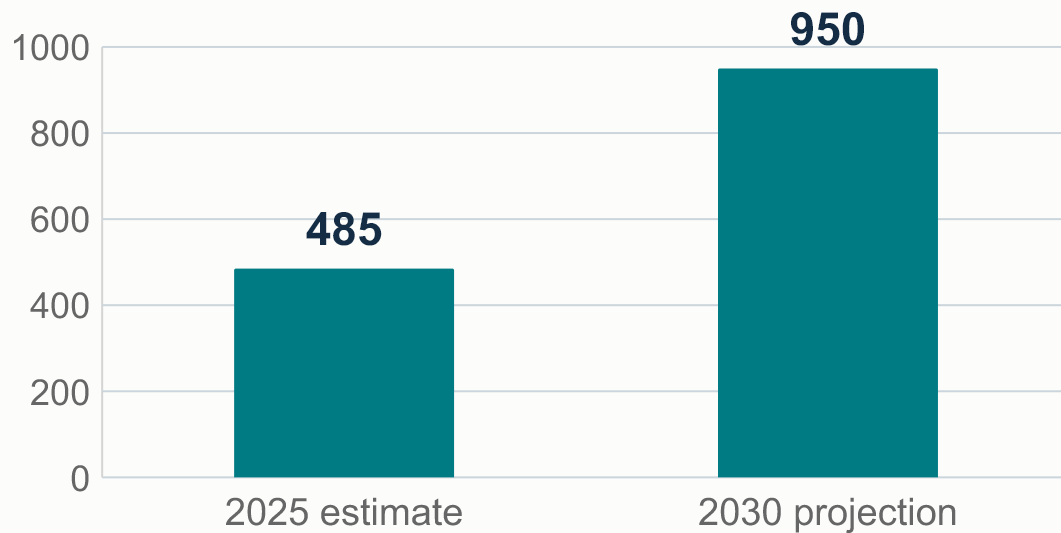
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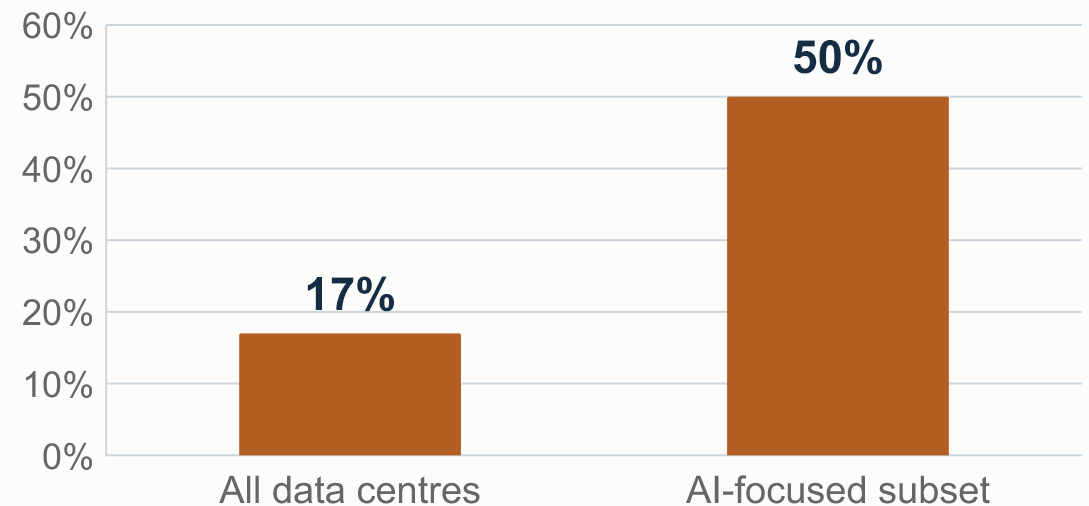
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Electricity demand and environmental costs

Global data-centre electricity (TWh/year)



Electricity growth during 2025

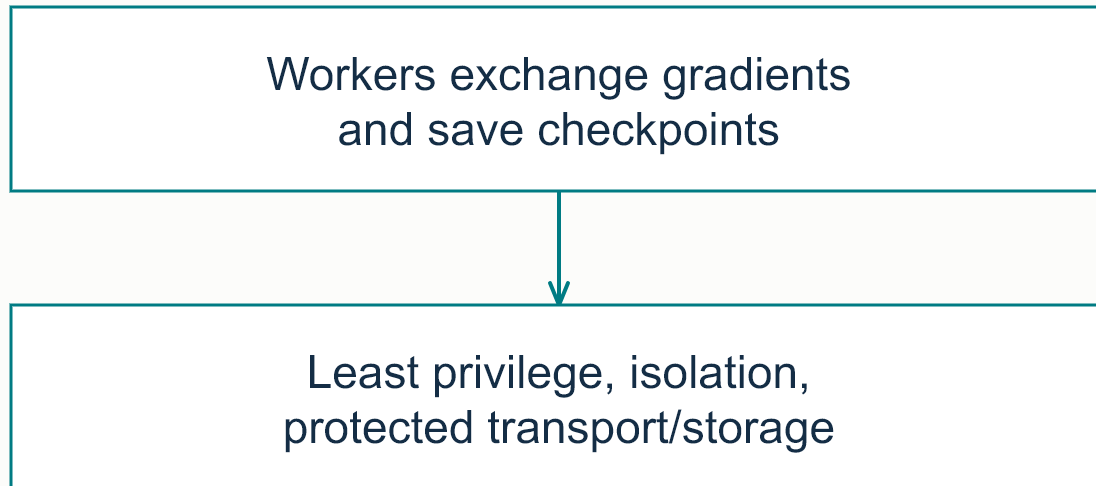


IEA 2026: ≈96% increase to 2030. These cover the whole sector, not AI training alone; growth categories overlap and cannot be added.

Research comparison: Green AI [10] emphasizes efficiency; water research [9] adds site/time constraints. Track energy, carbon and local cooling impacts.

Privacy, security and model fairness

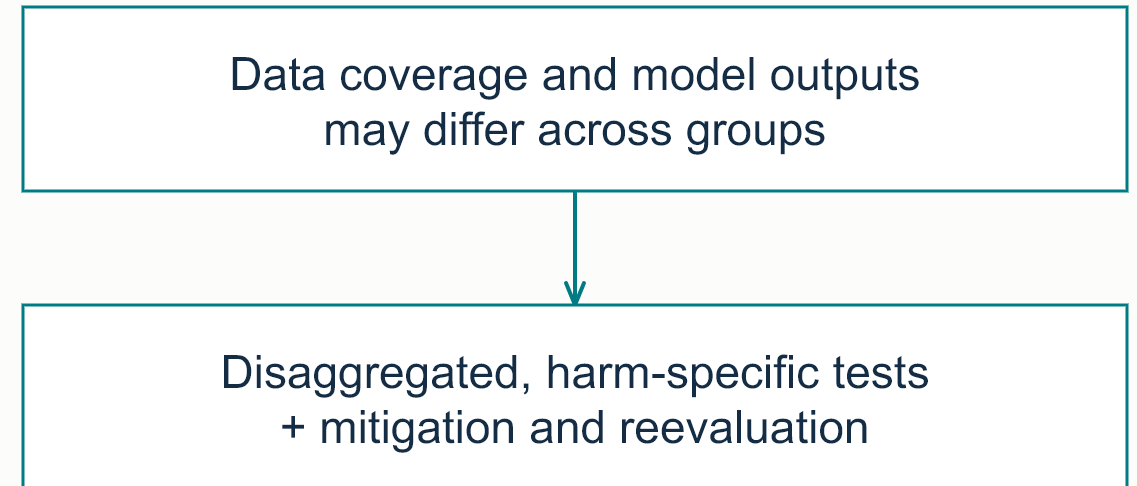
Security: protect distributed intermediates



Zhu et al. [11]: exposed gradients can reconstruct inputs in studied settings. Not every production collective leaks.

Provenance/permissions, replica access and dual-use review need owners. Encryption alone cannot establish fair outputs.

Fairness: evaluate affected groups



Gallegos et al. [12]: metrics, datasets and mitigations differ. Apply these to cluster-trained models; scale alone does not guarantee fairness.

Protect distributed data and evaluate harms; more compute is not a fairness guarantee.

Compute concentration limits participation

Evidence and consequences

Ahmed & Wahed [13]: institutional participation.
Strubell et al. [8]: workload cost. Different scopes reveal barriers to training and replication.

Historical studies do not establish current global access.

NAIRR: >600 teams and >6,000 students [7]; US reach does not establish worldwide equality.

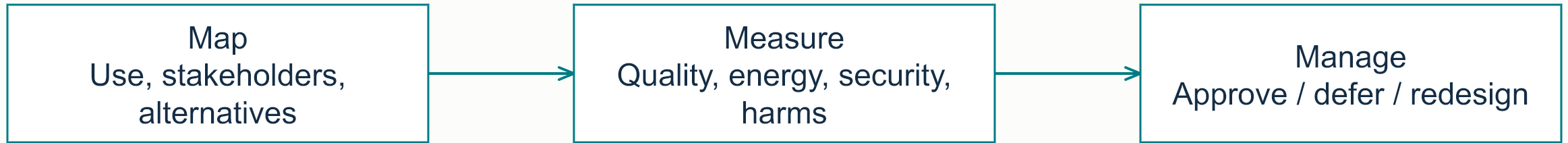
An equitable shared-cluster policy

Publish GPU-hour rules; reserve capacity for smaller institutions and replication.

Offer training and appeals; audit waiting times and awards with privacy safeguards.

Trade-off: reservations can lower utilization or delay urgent jobs. Prevent starvation.

Approve scale with evidence and ownership



Govern throughout: accountable owner, rights, data rules and equitable access

Green AI [10]: efficiency per useful outcome. OECD [15] + NIST [16]: rights, safeguards and review.

Carbon-aware scheduling [14]: shift flexible jobs within deadlines, water/site limits and equitable waiting commitments.

Record exceptions and outcomes; urgent work and forecast uncertainty can override deferral. Prevent starvation.

NIST Govern spans iterative review; unacceptable risks require redesign or rejection.

Scale useful work, then verify the outcome

Task 1: conditional performance

Independent inverse-gather pixels suit CUDA; locality, transfers and SM scheduling determine speedup.

GDS helps only a compatible storage-bound path.
Next: compile, compare CPU/GPU output and measure completed workloads.

Task 2: accountable scaling

Measure energy and useful output; protect data, evaluate group harms and allocate access transparently.

Next: test a real cluster job and review outcomes.
Sector projections and proposed policies are evidence limits, not measured results.

Detailed answers and supporting evidence

Task 1: implementation and analysis

Complete rotation and brightness examples, thread/grid calculations, host launch and stream completion.

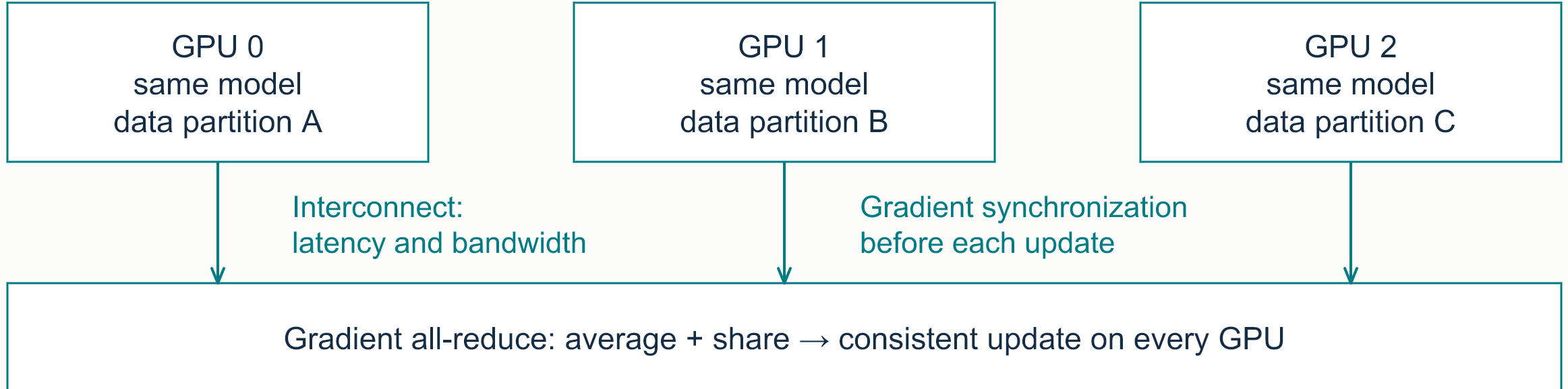
Worked rotation, coordinate assumptions, launch-shape trade-offs, Amdahl bound, profiling and GPUDirect Storage decision.

Task 2: research and policy

Parallel training, energy/carbon calculations, environmental accounting and research limitations.

Risk owners, fairness example, allocation policy, framework roles, carbon-aware scheduling and cross-topic comparisons.

Parallel training across a compute cluster



HPC combines processors, memory, networks and storage. Tensor/pipeline parallelism can also split a model.

More GPUs enable larger work; communication, serial work and imbalance limit speedup. Evaluate useful output against resource cost.

Scaling adds capacity and communication: ethical costs depend on useful work.

A faster job can consume more energy

Assumed configuration	Completed runtime	Whole-job energy
Lower parallel scale; 1 kW	2.0 hours	$1 \times 2.0 = 2.0 \text{ kWh}$
Higher parallel scale; 2 kW	1.2 hours	$2 \times 1.2 = 2.4 \text{ kWh}$

$E = \int P(t) dt \bullet \text{Carbon} \approx \sum E_{\text{interval}} \times \text{carbon intensity}$

Illustrative constant power assumptions, not measurements. Same useful output is assumed; faster completion uses 20% more energy here.

Environmental accounting and evidence limits

Define what is counted

Energy (kWh) integrates power (kW) over hours.
Operational emissions $\approx \sum$ energy in each interval $\times$ that interval's carbon intensity (kg CO₂e/kWh).

Report accelerators, host CPUs, storage/networking and allocated cooling overhead; include failed runs and repeated tuning when evaluating a project.

Hardware manufacture and disposal are embodied impacts beyond operational electricity. Water withdrawal and water consumption are different measures.

Compare the evidence

Strubell et al. quantify costs for selected historical NLP training/development workloads; those values are not universal 2026 model footprints.

Green AI motivates reporting efficiency alongside accuracy; Li et al. show why water varies with place/time. Carbon alone misses local water pressures.

IEA sector totals provide context, not causal attribution to one model. Forecasts depend on demand and infrastructure assumptions; local grid/cooling effects differ by site.

Governance controls and remaining risks

Risk / stakeholder	Proposed control	Residual limitation / owner
Dataset misuse / data subjects	Provenance, permission checks, retention limits and access logs	A licence alone may not justify every use; data steward reviews purpose.
Gradient/checkpoint exposure / participants	Least privilege, tenant isolation, protected transport and secure checkpoint storage	Authorized endpoints can still see sensitive values; security owner models insider risk.
Unsafe dual use / affected public	Authorized-use review, restricted access where justified and review of deployment risks	Controls cannot prevent every misuse; accountable owner weighs benefits/restrictions.
Distributed replicas / operators	Track copies, deletion obligations and incident response across workers	Replication increases the number of assets to protect; operator audits recovery.

These are proposed mitigations informed by research/frameworks, not a proof of privacy or a compliance certification.

Fairness evaluation with large-scale compute

Compute connection

An illustrative multilingual LLM service may have unequal performance or stereotyped outputs across language groups. Identify the actual harm before choosing a metric.

A cluster enables larger datasets and repeated training, but neither data volume nor GPU count guarantees representative data or equitable outcomes.

Reserve evaluation compute for affected groups; document dataset coverage and compare disaggregated error or harmful-output rates using a task-appropriate test.

Evidence and decision

Gallegos et al. separate metrics, evaluation datasets and mitigation stages. A single aggregate accuracy score can hide a relevant subgroup harm.

Specify justified acceptance criteria before training; examine uncertainty and small-group sample limits. Mitigate, reevaluate, and document unresolved harms.

Security and fairness have different objectives: encrypted gradients do not establish fair outputs. The source supports model-bias analysis, not a claim that HPC itself causes bias.

Unequal access to HPC resources

Evidence	What it shows	What it cannot establish
Ahmed & Wahed [13]	Historical research participation and compute concentration	Current worldwide access or a universal causal effect
Strubell et al. [8]	Selected NLP workload costs and financial barriers	The cost of every current AI workload

Current response: NAIRR supported >600 teams and >6,000 students

NSF, March 2026; US programme since 2024. Reach does not prove global equality.

Proposal: transparent GPU-hour allocation, smaller-institution support and appeals. Trade-off: reserved access can reduce immediate utilization or delay urgent jobs.

Shared capacity helps; allocation, expertise and reproducibility still need support.

Access research and allocation choices

Research comparison

Ahmed and Wahed study historical conference participation and a compute divide. Strubell et al. examine financial/resource barriers in selected NLP workloads.

The first concerns institutional participation; the second concerns workload cost. Together they motivate access support, but neither is a current worldwide access census.

NSF's March 2026 participation figures describe a US shared-resource response. Programme reach is not proof of equal allocation, global access or eliminated inequality.

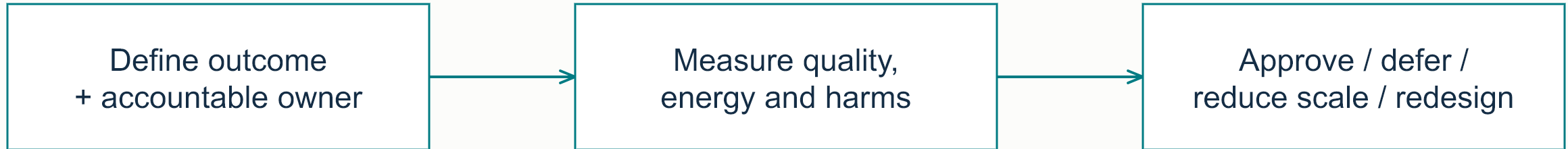
Proposed shared-cluster policy

Publish eligibility, GPU-hour limits and allocation criteria; reserve some capacity for teaching, smaller institutions and independent replication.

Provide application support and training; allow appeals. Track applicant-to-award outcomes and waiting times across institution groups with privacy safeguards.

Trade-offs: reservations may reduce immediate utilization; queueing can delay urgent work; unrestricted access can conflict with security/budgets. Prevent repeated deferral or resource starvation.

An accountable HPC approval process



Green AI: efficiency evidence [10] • Carbon-aware scheduling: flexible work [14]

OECD: values and rights [15] • NIST: Govern, Map, Measure, Manage [16]

Use an energy budget, security/group evaluation and transparent access rules. Record owner, rationale, exceptions and outcomes.

Measurement informs action: shift flexible jobs to cleaner periods within deadlines/capacity. Forecast uncertainty, water stress and urgent access can change the decision.

Approve useful work with explicit budgets, safeguards and an accountable owner.

Framework roles and accountable decisions

Function / evidence	Concrete action	Decision / responsible role
Govern / OECD values + NIST	Assign ownership, data rules, access policy and escalation	Accountable owner accepts obligations and review process.
Map / NIST	Define use, stakeholders, misuse risks, deadline and alternatives	Data/risk stewards establish the application-specific scope.
Measure / Green AI + NIST	Track energy per useful outcome, model quality, group harms and scaling efficiency	Operator/evaluator provides evidence and uncertainty.
Manage / scheduling research + NIST	Approve, defer, reduce scale or redesign; enforce safeguards and review outcomes	Owner authorizes exceptions or rejects unresolved risks.

Our proposal combines measurement, scheduling and risk management. NIST Govern spans the other functions; risk management is iterative, not a certification.

Carbon-aware scheduling under constraints



Research [14] demonstrates shifting temporally flexible workloads using forecasts while preserving daily capacity. It does not show every urgent GPU job can wait.

Proposed rule: defer only if deadline, energy budget, water/site limits and equitable waiting-time commitments remain satisfied; record exceptions.

No feasible window: reduce scale, redesign or approve a justified urgent exception. Prevent indefinite deferral. Data residency and transfer costs constrain moving work between sites.

Flexible jobs may move; urgent jobs, water constraints and access commitments can override deferral.

Conclusions and next measurements

GPU rotation

Inverse gathering exposes independent pixel work. SM scheduling, memory locality and transfer costs determine achieved performance.

Choose a CPU baseline and compare completed equivalent pipelines. GDS is conditional on storage bottlenecks, a compatible data path and verified support.

Next: compile the teaching examples, check output against a CPU reference and profile representative image sizes/angles/batches. No result is assumed.

Responsible scaling

More compute can enable valuable research while concentrating access and increasing resource demand. Approve scale for useful outcomes, with safeguards.

Combine job-level energy/resource evidence, separate security/fairness evaluation, transparent allocations and deadline-aware scheduling.

Next: measure a real cluster job's energy, quality, scaling efficiency and allocation outcomes; revise policy when evidence or conditions change.

Optional supporting material

Task 1: implementation expansions

The Task 1 answers already contain the rotation formula, worked calculation, kernel, host launch, launch-shape comparison and stream/overlap example.

The following slides expand coordinate conventions and complete source examples, then give an optional profiling plan and GDS background.

Task 2: supplementary comparisons

The Task 2 answers already contain energy/carbon calculations, research comparisons, governance and fairness decisions, access policy and accountable scheduling.

Additional parallelism and cross-topic comparison tables follow. Website references and one brief AI-use declaration complete the deck.

Parallelism, dependencies and scaling limits

Model	Partition and communication	Scaling limitation
Data parallel	Replicate model; partition batches; reduce/average gradients before a consistent update	Collective latency/bandwidth and stragglers.
Tensor parallel	Partition computations within a layer; exchange partial results	Frequent inter-device communication.
Pipeline parallel	Assign layers to stages; pass activations/gradients	Stage imbalance and pipeline bubbles.
Fixed workload	More GPUs reduce only the parallelizable portion	Serial work and communication bound speedup.
Larger workload	Scale capacity/model/data with added resources	A different claim from halving a fixed job's runtime.

All-reduce is a collective result-sharing concept, not a claim that every AI framework uses the exact classroom MPI code.

Research comparisons across all four topics

Topic	Comparison	Interpretation / limit
(a) Environment	Cost/efficiency research [8,10] versus water-footprint research [9]	Useful-output efficiency and local resource pressure need different measures.
(b) Governance	Gradient reconstruction [11] versus fairness evaluation survey [12]	Data exposure and unequal outcomes are distinct failure mechanisms.
(c) Access	Institutional participation [13] versus workload cost/equity [8]	Concentration and financial barriers reinforce a concern; scopes differ.
(d) Frameworks	Efficiency-reporting argument [10] versus operational scheduling [14]	Measurement motivates action; scheduling depends on flexibility and forecasts.

Current institutional evidence [6,7] and governance frameworks [15,16] complement the research; they do not replace its methodological limits.

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AI use declaration

AI model and tool

OpenAI GPT-6, accessed through Codex.

How AI was used

Research and source checks; drafting and revision of explanations and CUDA teaching examples; creation of diagrams and charts; presentation formatting and assembly.